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Faculty of Applied Computer Science

Data Visualization (Final Portfolio)

**Air Pollution and Respiratory Mortality across the
German Federal States, 2019 to 2023**

A Regional Ecological Visualization Study

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Abstract

Outdoor air pollution and respiratory disease remain among the leading public health concerns in Europe. Working as data analyst consultants, we set out to test whether two regulated pollutants, particulate matter (PM₁₀) and nitrogen dioxide (NO₂), are associated with respiratory mortality (ICD-10 codes J00 to J99) across the sixteen German federal states between 2019 and 2023. The study brings together official air quality data from the German Environment Agency and official mortality data from the Federal Health Monitoring system. Our central result is that the conclusion depends entirely on how mortality is measured. Expressed as raw death counts, pollution appears harmful, but that impression simply tracks population size. Expressed as a crude population rate, the association turns negative, because the oldest and most rural states carry the highest respiratory mortality while experiencing the least pollution. Only after age standardisation, which strips out the effect of differing age structures, does a genuine signal appear. Nitrogen dioxide is then positively and significantly associated with respiratory mortality, with a correlation of 0.31 ($p = 0.005$) across all observations and 0.50 across the state averages, while particulate matter shows no clear relationship. The hypothesis therefore holds for nitrogen dioxide once the analysis accounts for age. We argue that this progression through three measures is precisely the kind of confounder-aware rigour that a credible regional health analytics service has to offer.

Keywords: air pollution, respiratory mortality, ecological study, data visualization, Germany, PM₁₀, NO₂, age standardisation, confounding.

Abbreviations: UBA, Umweltbundesamt (German Environment Agency); GBE, Gesundheitsberichterstattung des Bundes (Federal Health Monitoring); WHO, World Health Organization; ICD-10, International Classification of Diseases, tenth revision; PM₁₀, particulate matter below ten micrometres; NO₂, nitrogen dioxide; ASR, age-standardised rate; ESP, European Standard Population; AQG, Air Quality Guideline; CI, confidence interval.

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1. Introduction

Air pollution is the single largest environmental risk to human health. The World Health Organization attributes roughly 4.2 million premature deaths each year to ambient outdoor air pollution, and the respiratory and cardiovascular systems are its principal targets (WHO, 2024). Respiratory disease on its own accounts for about 363,500 deaths every year in the European Union, close to seven percent of all deaths (Eurostat, 2024). For a life sciences firm looking to enter the digital health and environmental monitoring market, the meeting point of environment and respiratory health is an attractive place to build. The data are plentiful, they vary from region to region, and they speak directly to prevention, to insurers, and to public sector clients.

We have written this report from the position of an internal research and development team making the case for a new service line, a regional respiratory risk analytics platform. To justify the investment we need to show two things. First, that the raw data can be turned into clear and decision ready insight. Second, and just as important, that arriving at the correct insight takes methodological care that ready made tools do not provide. Germany is a good place to prove the point, because it publishes high quality and openly licensed air quality and mortality data at the level of its sixteen federal states. That allows a properly regional analysis across both space and time. We use Germany over the period 2019 to 2023 as a worked demonstration of the product.

2. Problem definition and state of the art

Particulate matter and nitrogen dioxide are among the pollutants of greatest public health concern. Both come mainly from combustion, whether from road traffic, heating, or industry, and both are firmly linked to airway inflammation, reduced lung function, and worsening of chronic respiratory disease (WHO, 2021). In its 2021 global update the World Health Organization tightened the recommended annual mean guideline values to 15 micrograms per cubic metre for PM10 and 10 micrograms per cubic metre for NO2 (WHO, 2021). As the results below show, most German states still sit above the NO2 guideline, so the question of exposure is a real one.

At the level of the individual, the evidence is strong. Long term cohort studies across Europe and North America connect both particulate matter and nitrogen dioxide to reduced lung function, chronic obstructive pulmonary disease, worsening asthma, and higher respiratory mortality, and nitrogen dioxide in particular serves as a reliable marker for the mixture of traffic pollutants found in cities (WHO, 2021). The question this project addresses is whether that individual level signal is still visible in aggregate, regionally averaged data, and if it is, what it takes methodologically to see it rather than lose it to confounding.

The difficulty is that the most accessible health and exposure data are aggregated to administrative regions rather than measured on individuals. Studies built on such data are ecological, and they are open to the ecological fallacy, the error of reading individual level relationships off group level correlations (Greenland & Robins, 1994). A common and damaging version of this error is to relate pollution to unadjusted death counts, or to crude rates that take no account of the age structure of the population. Because respiratory deaths fall so heavily on older people, any region with an older population will record high respiratory mortality whatever its air quality. Age standardisation, which reweights each region's death rates onto a shared reference age distribution, is the established remedy (Ahmad et al., 2001; Eurostat, 2013). The state of the art, then, is not to plot pollution against deaths and stop there. It is to normalise both exposure and outcome correctly, to take out known confounders such as age, and to report the remaining uncertainty honestly. That is the methodological value we set out to demonstrate.

3. Objectives

The work has four objectives. The first is to bring the official German pollution and mortality data together into a single, reproducible table organised by state and year. The second is to describe the spatial and temporal patterns of PM10, NO2, and respiratory mortality across the period using a varied set of advanced visualizations. The third is to test the team's hypothesis, that higher long term exposure to PM10 and NO2 goes with higher respiratory mortality, using suitable descriptive statistics and correlation analysis, and in doing so to show how the choice of mortality measure changes the answer. The fourth is to turn the findings into a defensible recommendation for the firm.

4. Methods

Data sources. Annual mean concentrations of PM10 and NO2 for every monitoring station were taken from the German Environment Agency air data service for the years 2019 to 2023 (UBA, 2024). Respiratory mortality was taken from the Federal Health Monitoring system as deaths coded to ICD-10 J00 to J99, by federal state and year, in three forms: absolute counts, the crude rate per 100,000, and the age-standardised rate per 100,000 based on the European Standard Population of 2013 (GBE-Bund, 2024). State resident populations were used to reconstruct counts where needed.

Processing. Station level pollutant readings were averaged, without weighting, into a state mean for each year, and federal monitoring network rows and flagged values were left out. Mortality figures were extracted for the J00 to J99 group and reshaped to one row per state and year. The sources were joined on state and year to give a tidy table of sixteen states over five years, that is eighty observations, each carrying PM10, NO2, and the three mortality measures. The primary outcome is the age-standardised rate; the counts and crude rates are kept in order to show what the choice of measure does. Age standardisation was carried out by the Federal Health Monitoring system using the direct method against the European Standard Population of 2013 (Eurostat, 2013), in which each state's age specific death rates are reweighted onto a common reference age distribution, so that a state no longer looks worse simply because its population is older. This is the decisive processing step, because Germany's federal states differ a great deal in age structure. The eastern states and the rural north are substantially older than the southern city regions, and that one difference dominates any naive comparison of crude rates.

Analysis. We report descriptive statistics for every variable. Associations between each pollutant and each mortality measure were quantified with Pearson correlations, with 95 percent confidence intervals from Fisher's z transformation, and checked with Spearman's rank correlation. We computed the pooled correlation across all eighty observations and the correlation of the five year state means, which isolates the purely spatial signal. A multiple linear regression of age-standardised mortality on PM10 and NO2 gauged their joint contribution. Throughout, we read the findings as associations only; the ecological design does not support claims about cause.

Visualization. All graphics use colour blind safe palettes and cover several chart families: a three panel comparison of the measures, scatter plots with regression lines and confidence bands, a dual axis time series, tile grid choropleths, box plots, a co-exposure bubble matrix, and a connected dot change plot, together with an interactive station level dashboard described in the appendix.

5. Results

5.1 The measurement decision that changes the answer

The most important result is a methodological one. Figure 1 relates NO₂ to respiratory mortality using the same data under three measures. With raw counts the relationship is positive, at a correlation of 0.44, but counts move almost perfectly with population, at a correlation of 0.98, so this really only says that larger, more urban states have both more deaths and more traffic. With the crude rate per 100,000 the association turns negative, at minus 0.27, because older and low pollution rural states carry high crude mortality. Only with the age-standardised rate, which lifts the age confounder out of the comparison, does a positive association appear, at 0.31. It is the measure, not the data, that sets the conclusion, which is a decisive lesson for any analytics product. It is worth being clear about why each measure fails. Counts answer the question of where the most people die, which is largely a map of population. Crude rates answer what share of each population dies, which in an ageing country is driven by how old each region happens to be. Only the age-standardised rate answers the question we actually care about, namely where respiratory mortality is highest once populations are placed on the same age footing, and once the question is put that way, nitrogen dioxide emerges as a coherent predictor.

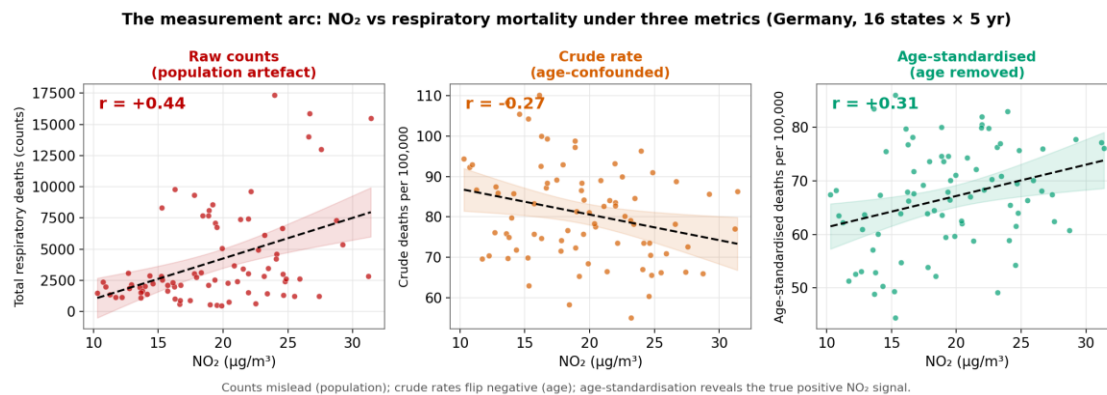


Figure 1. Nitrogen dioxide against respiratory mortality under three measures. Counts mislead through population, crude rates turn negative through age, and age standardisation reveals the true positive signal.

5.2 Distribution of exposure and outcome

Table 1 summarises the corrected data. PM₁₀ averaged 15.1 micrograms per cubic metre, with a standard deviation of 1.8 and a range from 11.6 to 19.6. NO₂ averaged 19.5 micrograms per cubic metre, with a standard deviation of 5.1 and a range from 10.3 to 31.4, close to twice the guideline value of 10 that most states exceed. Age-standardised respiratory mortality averaged 66.9 per 100,000, with a standard deviation of 9.6 and a range from 44.4 to 86.0. It is both lower and less spread out than the crude rate, whose mean was 80.9, which reflects Germany's older than reference population.

Table 1. Descriptive statistics, sixteen states over five years (n = 80).

Variable	Mean	SD	Min	Median	Max
PM10 ($\mu\text{g}/\text{m}^3$)	15.1	1.8	11.6	14.8	19.6
NO2 ($\mu\text{g}/\text{m}^3$)	19.5	5.1	10.3	19.3	31.4
Crude deaths per 100,000	80.9	11.9	54.7	79.6	109.9
Age-standardised deaths per 100,000	66.9	9.6	44.4	66.8	86.0

Figure 2 shows that both pollutants fell over the period, PM10 by about a fifth and NO2 by about a quarter between 2019 and 2023, and that their spread across states narrowed, a broad improvement in air quality nationwide.

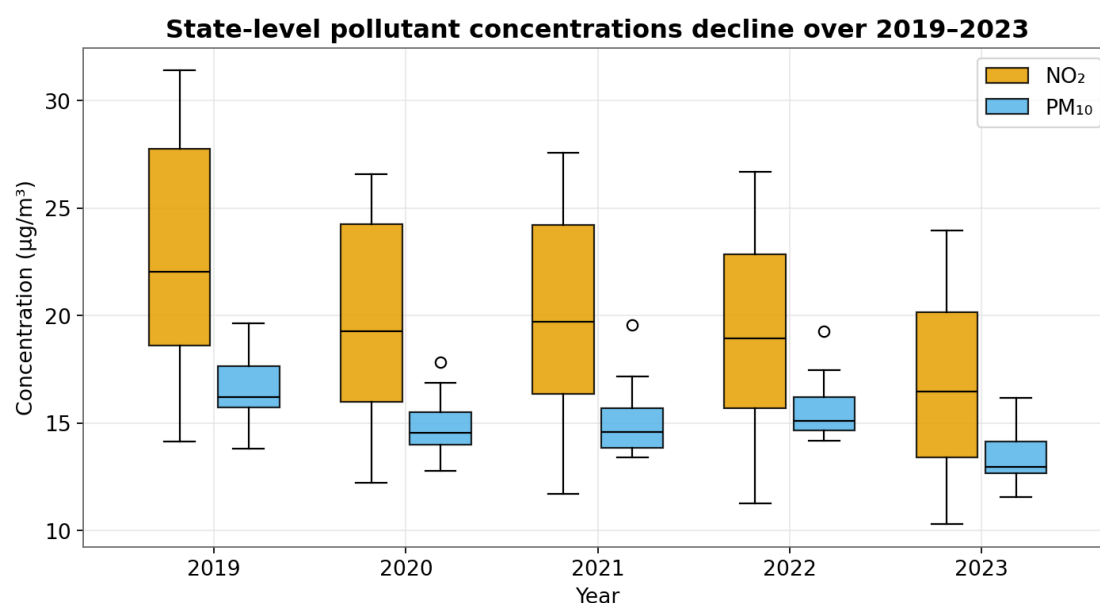


Figure 2. State level distributions of nitrogen dioxide and particulate matter by year. Both pollutants decline and converge across 2019 to 2023.

5.3 The pollution and mortality association

With the age-standardised outcome the hypothesis holds for nitrogen dioxide. Figure 3 shows a clear positive slope for NO2 and a flat one for PM10, and Table 2 puts numbers on them. Pooled across all eighty observations, NO2 correlates with age-standardised mortality at 0.31, with a 95 percent confidence interval from 0.10 to 0.50 and a p value of 0.005, while PM10 shows only a weak and non significant positive relationship at 0.11. The NO2 association survives the sterner test as well. Across the sixteen five year state means, which removes the repeated measurement of the same states over the years, it is 0.50 with a p value of 0.049, so the spatial signal is real and not a by-product of pooling. A joint regression of age-standardised mortality on both pollutants returns a significant NO2 coefficient, at plus 0.67 per microgram per cubic metre with a p value of 0.007, and a non significant PM10 coefficient, at minus 0.46, together explaining about a tenth of the variance. PM10 and NO2 are themselves moderately correlated, at 0.55, so they carry overlapping information about exposure, and nitrogen dioxide, as a direct marker of traffic combustion, is the stronger of the two.

Table 2. Correlations with age-standardised respiratory mortality.

Comparison	Pearson r	95% CI	p value	Spearman
NO2, pooled (n = 80)	+0.31	+0.10, +0.50	0.005	+0.31
PM10, pooled (n = 80)	+0.11	-0.11, +0.32	0.33	+0.06
NO2, state means (n = 16)	+0.50	+0.00, +0.80	0.049	n/a
PM10, state means (n = 16)	+0.28	-0.25, +0.68	0.30	n/a
NO2 versus crude rate	-0.27	n/a	0.014	n/a
NO2 versus raw counts	+0.44	n/a	<0.001	n/a

After age-standardisation, NO₂ is positively associated with respiratory mortality

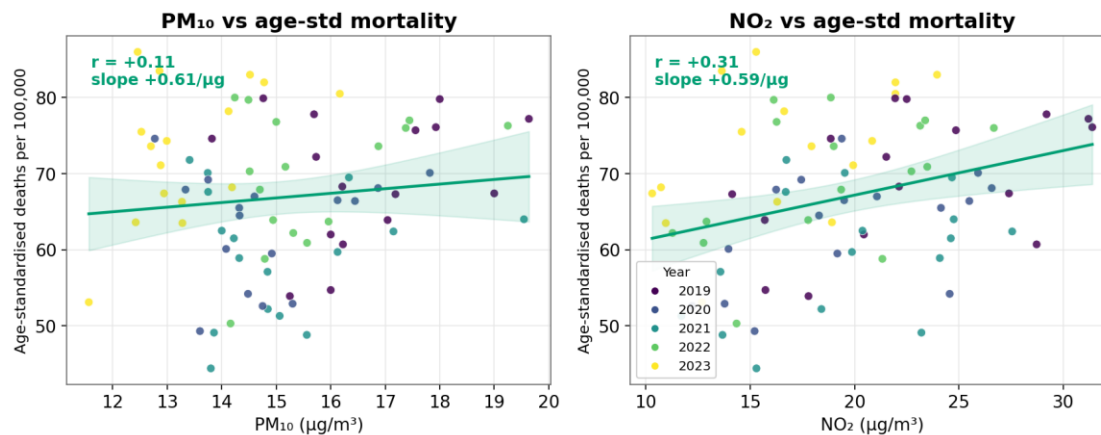


Figure 3. Age-standardised respiratory mortality against particulate matter and nitrogen dioxide, all sixteen states over five years, with regression line and 95 percent confidence band. Points are coloured by year.

5.4 Where the mortality is, and how the map realigns

Figure 4 sets age-standardised mortality and NO₂ side by side as tile grid maps. Under the crude rate the two had looked like mirror images. After age standardisation they line up in the urban states. Berlin, North Rhine-Westphalia, and Hamburg, all dense, traffic heavy, and high in NO₂, now sit among the highest for age-standardised mortality, at between 73 and 74 per 100,000, and that is where the positive correlation comes from. The rural states of Lower Saxony and Thuringia stay high, a reminder that other factors matter too, while the clear low outlier is Baden-Württemberg, high in NO₂ yet lowest in mortality, in keeping with its younger and wealthier population. The re-ranking is striking in its detail. Berlin climbs from the middle of the crude distribution to third once age is removed, while Saxony and Saxony-Anhalt, old, eastern, and relatively clean, fall to the bottom. That movement is the visible fingerprint of standardisation: the states whose high crude mortality was only an ageing effect drop away, and the urban, traffic exposed states are left at the top. What remains is a real, if imperfect, spatial correspondence between nitrogen dioxide and respiratory mortality.

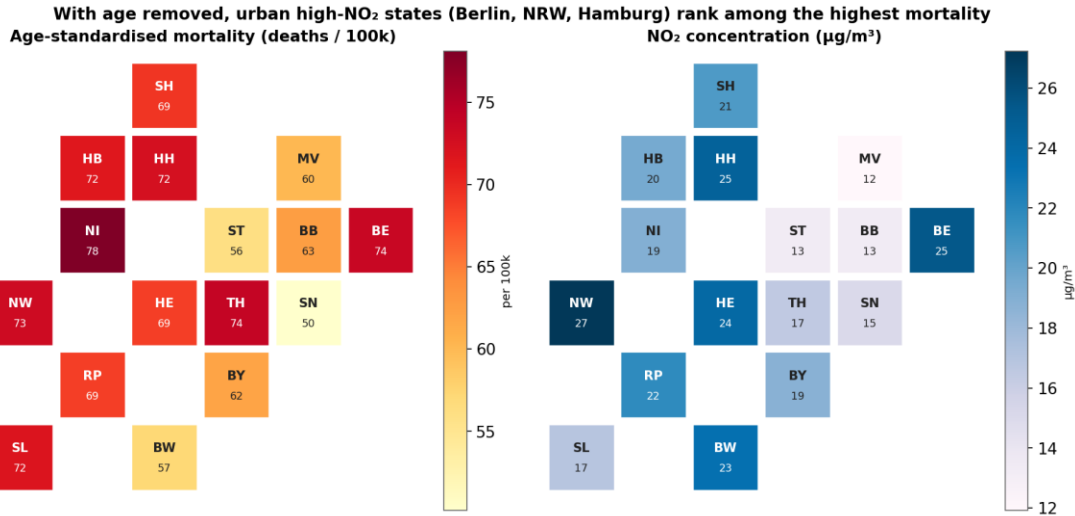


Figure 4. Tile grid maps of five year mean age-standardised mortality on the left and nitrogen dioxide on the right. Urban high NO₂ states now rank among the highest for mortality.

The co-exposure matrix in Figure 5 tells the same story for 2023. The larger and darker bubbles, which mark the states with the highest age-standardised mortality, now sit mainly in the high NO₂ band, the reverse of what the crude rate had suggested.

Co-exposure matrix (2023): larger age-standardised-mortality bubbles sit in the high-NO₂ band

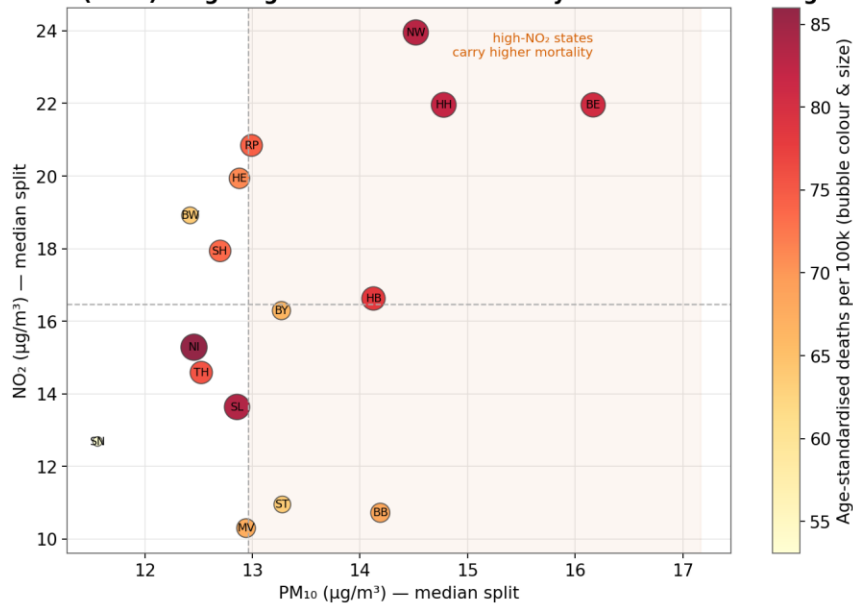


Figure 5. Co-exposure risk matrix for 2023. Bubble size and colour show age-standardised mortality, which is raised in the high NO₂ states.

5.5 The temporal picture and COVID-19

Figure 6 places the national trends together. Age-standardised respiratory mortality fell from about 69 per 100,000 in 2019 to a low of 59 in 2021, then rebounded to 73 by 2023.

The dip through 2020 and 2021 coincides with the COVID-19 lockdowns, when distancing measures sharply cut the circulation of influenza and other respiratory infections, and the rebound reflects their return. Pollution meanwhile fell steadily throughout. This is why cross sectional comparisons between states, rather than a naive pooling of years, carry the cleaner exposure signal. Figure 7 confirms at the state level that age-standardised mortality was higher in 2023 than in 2019 in almost every state.

Both pollution and (age-standardised) mortality dip during COVID; mortality rebounds by 2023

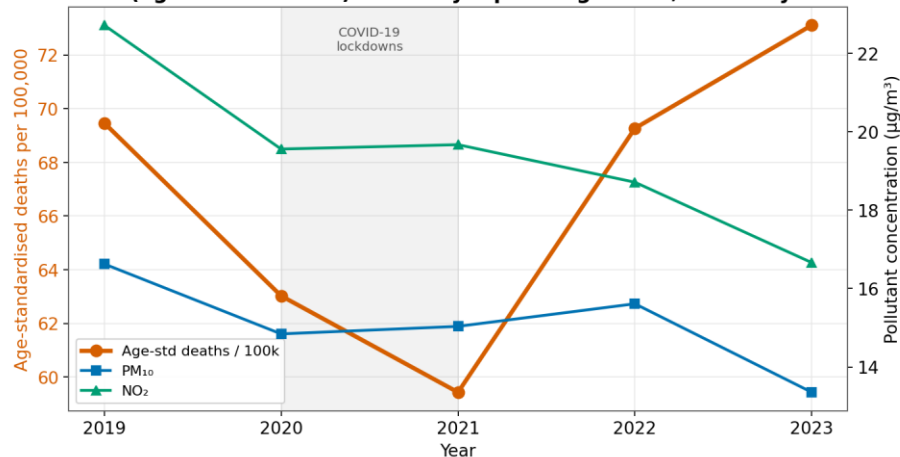


Figure 6. National mean trends. Age-standardised mortality dips through the COVID-19 lockdowns and rebounds, while pollution falls steadily.

Post-COVID rebound: age-standardised respiratory mortality rose in most states (2019 → 2023)

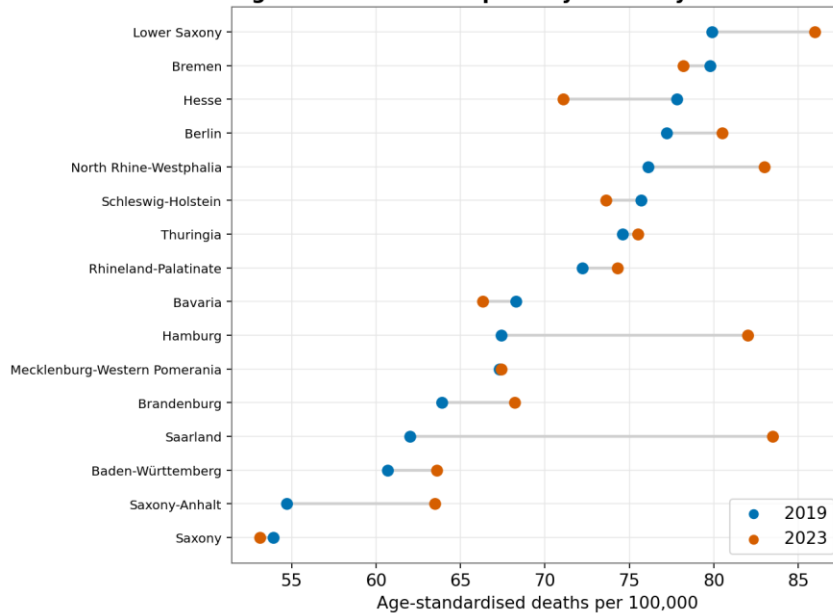


Figure 7. Change in age-standardised respiratory mortality from 2019 to 2023, by state.

6. Discussion

The headline finding is that the team's hypothesis holds, but only for nitrogen dioxide, and only once the analysis is done properly. Three points are worth drawing out.

The first is that the measure is decisive. In favour of the count and crude rate charts, they are simple and need no standardisation machinery. Against them, they turn the conclusion the wrong way. Counts manufacture a positive association out of population size, and crude rates manufacture a negative one out of age structure, as Figure 1 makes plain. Neither is the real signal. Age standardisation, by reweighting every state onto the European Standard Population of 2013, removes the age confounder and brings out a positive NO₂ association that is consistent across the pooled data, the state mean spatial analysis, and the regression. For a commercial product the lesson is central: credibility rests on getting the denominator and the confounders right.

The second point concerns why nitrogen dioxide and not particulate matter. Nitrogen dioxide is a direct marker of local traffic combustion and follows the urban gradient closely, so contrasts in NO₂ between states are large and meaningful. PM₁₀ is more uniform across space, with a standard deviation of 1.8 against 5.1 for NO₂, and it has more varied sources, which leaves it little power to separate states. Its flat and non significant relationship here should not be read as evidence of safety, only as limited ecological contrast. The result sits comfortably with individual level cohort evidence, which ties both pollutants to respiratory harm (WHO, 2021). For the proposed service the practical implication is concrete. A regional risk product has to treat age structured denominators as a first class input rather than an afterthought, and it should lead with nitrogen dioxide as its traffic exposure signal while carrying PM₁₀, and in time PM_{2.5} and ozone, alongside it. Exposure inputs deserve the same care, since the station level dashboard in the appendix shows that within a single state the roadside traffic monitors read far higher than the background sites, so a population weighted exposure estimate would sharpen the unweighted state means used here.

The third point is that this remains an ecological association. A significant regional correlation is not proof of causation for individuals. Other differences between states, such as smoking, occupational and indoor exposure, healthcare access, and urbanisation, move together with NO₂ and cannot be separated here. Exposure is a state level annual mean that smooths over large variation within states and across seasons. The five year window is short and shaped by the pandemic, and with only sixteen spatial units the state mean test, while significant, has limited power. These are the natural next steps for the product: rates broken down by age and sex, socioeconomic adjustment, station weighting, a wider set of pollutants, and seasonal exposure windows.

7. Conclusions

Across Germany's sixteen federal states over 2019 to 2023, nitrogen dioxide is positively and significantly associated with age-standardised respiratory mortality, at 0.31 across all observations, 0.50 across the state means, and with a regression coefficient significant at a p value of 0.007, while particulate matter shows no clear ecological relationship. The team's original hypothesis therefore holds for nitrogen dioxide, but only after proper age standardisation. The count and crude rate versions of the same data point in the wrong or the opposite direction. No claim about cause can be made from these aggregate figures.

For the firm this is a persuasive proof of concept. It shows, on real data, that a regional respiratory risk service earns its value exactly where off the shelf pollution dashboards fail. Counts and crude rates would have shipped either a spurious or an inverted conclusion, whereas a confounder-aware, age-standardised analysis recovers the genuine nitrogen dioxide signal and states its uncertainty openly. We recommend that the research and development department resource a second iteration, with rates broken down by age and sex and adjusted for socioeconomic factors, a wider pollutant set, and the interactive monitoring dashboard, to turn this proof of concept into a client ready product.

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Appendix

Appendix A. Source code and reproducibility. The full analysis is provided as the R Markdown notebook `analysis.Rmd`. It rebuilds the master dataset from the raw files and regenerates every figure in R using `tidyverse`, `ggplot2`, `plotly`, and `leaflet`. To reproduce it, place the data files in the working folder, set the `data_dir` path, install the listed packages, and knit the notebook.

Appendix B. Interactive visualizations and maps. The file `Deep_Dive_Dashboard.html` is a linked dashboard built with `crosstalk` and `plotly` over the complete station level PM10 and NO2 data, with filters for state and year and stations coloured by type, whether traffic, background, or industry. The interactive Leaflet choropleth map of respiratory mortality and the linked plotly scatter are included in `analysis.Rmd` and render when the interactive chunks are set to `eval = TRUE`.

Appendix C. Data sources. Air quality data are the Umweltbundesamt annual station tabulations of PM10 and NO2 for the years 2019 to 2023. Mortality data are from the Gesundheitsberichterstattung des Bundes, covering respiratory deaths under ICD-10 codes J00 to J99 together with the age-standardised rates on the European Standard Population of 2013, for the years 2019 to 2023.

Appendix D. Data dictionary for `master_clean.csv`. The file holds one row per state and year, with the columns `State`, `Year`, `Deaths` (the respiratory death count for J00 to J99), `NO2` and `PM10` (the annual mean concentration in micrograms per cubic metre), `Crude_100k` (crude deaths per 100,000), and `AgeStd_100k` (age-standardised deaths per 100,000).

Appendix E. Figures and supporting files. The report includes seven figures, listed in the List of Figures. The supporting files are the cleaned dataset `master_clean.csv`, the fifteen minute presentation deck `Air_Pollution_Presentation.pptx`, and the styling file `report_header.html` used when knitting the notebook.